S Sudeep

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Professional Summary

Aspiring Data Engineer with expertise in Python, PySpark, SQL, Azure Databricks, Delta Lake, and Azure Data Factory. Hands-on experience building scalable Medallion Lakehouses across 33M+ records, designing Kimball Star Schemas, and applying Delta Lake storage optimizations including Z-Ordering and Broadcast Joins. Skilled in developing production-grade ELT/ETL pipelines and enabling sub-second natural language analytics using Databricks Genie.

Education

Sri Venkateswara College of Engineering, Tirupati — Tirupati, India
B.Tech in Computer Science and Engineering (CGPA: 8.78/10.0) — 2023 – 2027

Sreenivasa Junior College, Kalikiri — Kalikiri, India
Intermediate (MPC) (Percentage: 92.8%) — 2021 – 2023

Technical Skills

Languages and Querying: Python, SQL (Window Functions, CTEs, Joins, Aggregations), PySpark, Spark SQL
Big Data and Storage: Apache Spark, Delta Lake, ACID Transactions, Schema Enforcement and Evolution, Data Skipping
Cloud and Orchestration: Azure Databricks, Azure Data Factory (ADF), ADLS Gen2, Azure SQL
Data Modeling: Medallion Architecture (Bronze/Silver/Gold), Kimball Star Schema, Fact and Dimension Modeling, SCD Type 1
Pipeline Engineering: Incremental Loading (Watermark Pattern), Delta Z-Ordering, Broadcast Hash Joins, Batch ELT/ETL
Tools and Platforms: Git, GitHub, Databricks Genie (AI/BI), Unity Catalog

Projects

Enterprise Medallion Lakehouse and AI Analytics (33.8M Records) | GitHub Repo

- Built an end-to-end Medallion Lakehouse pipeline ingesting and standardizing 33.8M+ raw transactional records into Delta tables.
- Engineered Silver transformations enforcing schema constraints, deduplicating 206K unique customers, and unioning datasets.
- Designed a Kimball Star Schema with a dual-grain fact architecture, isolating order checkouts (`fact_orders`, 3.34M rows) from line items (`fact_order_items`, 33.8M rows) to reduce high-level query scan costs by 90%.
- Optimized Delta storage using `ZORDER BY (user_id, product_id)` and applied Broadcast Hash Joins on dimensional lookups.
- Configured Databricks Genie with semantic routing, enabling non-technical stakeholders to execute sub-second natural language queries to Spark SQL.

End-to-End Azure Data Pipeline for Car Sales Analytics | GitHub Repo

- Designed an automated cloud ELT framework orchestrating multi-source relational ingestion into ADLS Gen2 for analytical reporting.
- Implemented an incremental loading mechanism in ADF leveraging dynamic watermark tables, stored procedures, and scheduled triggers.
- Developed PySpark transformation routines enforcing schema validation, record deduplication, null imputation, and relational joins.
- Modeled dimensional Gold tables using SCD Type 1 logic and surrogate key assignments to maintain clean and reliable reporting models.

Achievements and Certifications

- **LeetCode:** Solved 180+ problems covering core Data Structures, Algorithms, and advanced SQL query patterns.
- **LeetCode Data Navigator Badge:** Earned for solving multi-table relational queries, joins, aggregations, and window functions in SQL.
- **IEEE Student Member:** Active participant in technical workshops and engineering student chapters.